From Reconstruction to Anticipation

Predicting future tissue geometry from monocular laparoscopic video

FYP-2 Mid Evaluation · Modules 5 and 6

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What the project is for

A surgeon operating through a single laparoscope sees a flat image of the present moment. Our goal is a system that reconstructs the scene in 3D and then anticipates how the tissue will deform next — from that one camera, with no depth sensor and no clinical annotation.

O1 · Reconstruct
4D scene geometry from monocular laparoscopic video, without stereo input or labels.

COMPLETE · FYP-1

O2 · Learn deformation
A self-supervised model of how tissue moves, trained on the reconstructed geometry.

COMPLETE · MODULE 5

O3 · Predict ahead
Future tissue geometry at 1–3 frames, with a calibrated estimate of confidence.

IMPLEMENTED · MODULE 6

O4 · Evaluate and position
Against classical and learned baselines, and against the published literature.

IN PROGRESS · M7–M8

Research question

Can future 3D tissue geometry be predicted from monocular surgical video alone — and can the system say how much to trust each prediction?

This is a research FYP. The deliverable is a methodology, a set of measured results, and a comparison against the literature — not a product.

Faculty feedback from FYP-1, and what we did about it

Implement a base paper for comparative analysis	→	**Two base papers implemented** PointRNN (Fan & Yang) for point-cloud forecasting; Luo & Hu (CVPR 2021) for point diffusion. *module7_baselines.py*
Select a few SOTA papers and implement them	→	**Eleven methods on one protocol** Five classical and five learned predictors, plus ours, all scored on identical windows. *module7_baselines.py · 781 lines*
Do a detailed analysis — hyperparameter tuning, optimisation	→	**Four tuning studies** Inference cost against accuracy, context length, optimiser sensitivity, and diffusion ablations. *module8_hpo.py · 593 lines*
Start writing a paper in IEEE or Springer format	→	**IEEE conference manuscript drafted** Six pages, 22 references, tables generated directly from the result files. *paper/main.tex · paper/main.pdf*

All four are implemented and in the repository. Two still need their compute runs, which is item 1 on the remaining-work slide.

The methodology, finalised

Given the last 4 frames of 512 tracked tissue points, predict their displacement k frames ahead, for k ∈ {1, 2, 3}. No clinical labels at any stage.

Model

Conditional denoising diffusion
v-prediction, cosine schedule, T = 500
Min-SNR-γ loss weighting
Temporal transformer + two-way cross-attention · 16.3 M parameters

What it predicts

The correction to constant-velocity extrapolation, not the displacement itself — so a degenerate model falls back to an estimator 41% stronger than the naive baseline.

How it is queried

DDIM sampler, 20 steps
Direct (k ≤ 3), rollout (10 frames), and 5 independent samples whose spread is the uncertainty estimate.

Data and protocol

10 reconstructed videos · 512 anchor points · window W = 4
5,910 training windows, 1,461 validation windows
Self-supervised throughout — the targets come from the reconstruction, never from a human annotator.

Reported where it can be checked

The method is written up in full in the IEEE manuscript, Section III, with every hyperparameter stated. Each number in the paper is generated from a result file rather than transcribed by hand.

Every reported metric is inverted back into physical units before it is printed, so nothing on these slides is in normalised space.

Metrics chosen, and what they measured

Why these metrics

Chamfer Distance — correspondence-free, so it does not assume point identity holds across frames. Primary accuracy metric.
Mean Point Error — correspondence-aware, reported alongside it.
Directional cosine — separates "learned nothing" from "learned the direction but the wrong distance". Chamfer Distance alone cannot.
Spread–error correlation — tests whether uncertainty is calibrated.
PSNR / SSIM — image space, against the reconstruction ceiling.

Six quality tests, run as a suite

QT0 verifies the evaluation data is the data the model trained on, and runs as a gate before anything is predicted. QT1–QT5 cover short-horizon accuracy, rollout growth, diversity, smoothness and cross-scene generalisation.

What we measured

Accuracy — not yet ahead

CD 0.01568 against a naive baseline of 0.01557.
Constant velocity beats both, on all 10 videos.

Direction — clearly learned

Cosine +0.57 at t+1, +0.40 at t+2, +0.26 at t+3.
A model that learned nothing would read 0.00.
So the failure is magnitude, not direction.

Uncertainty — calibrated

Spread across 5 sampled futures correlates with realised error at r = 0.69. No baseline in the comparison provides this at all.

Two explanations were tested and eliminated before we drew any conclusion

- Error does not grow with distance from the training frames (ρ = +0.02, p = 0.61)
- Dropped preprocessing frames do not explain it (333 origins checked, 0 drops found)

How we compare with published work

3D reconstruction — base paper family

Method	Input	PSNR
LerPlane (MICCAI 23)	stereo	29.46
EndoGaussian (24)	stereo	30.25
Deform3DGS (MICCAI 24)	stereo	30.48
Ours	**monocular**	**30.10**

Indicative, not controlled: published figures use stereo supervision on different sequences. We state this in the paper.

What the left table establishes

Our reconstruction sits within 0.4 dB of the stereo state of the art while using a single camera and no depth sensor.

That geometry is the input Module 5 learns to predict.

Prediction — base paper family

PointRNN (Fan & Yang, 2019)

The closest published method: future point clouds from past ones, scored with Chamfer Distance. Implemented faithfully as our primary learned baseline.

Luo & Hu (CVPR 2021)

First diffusion model over 3D point clouds. Our denoiser is a conditional, temporally-grounded instance of it.

Full comparison in progress

11 methods on one protocol: 5 classical, 5 learned, ours. Module 7 implemented; the run is the next step.

Remaining work until the final evaluation

1. **Run the 11-method comparison and the HPO studies**
   Modules 7 and 8 are written and tested. Only the compute runs remain.
2. **Evaluate the two corrections already implemented**
   Magnitude calibration needs no retraining; the residual target needs one retrain.
3. **Move to leave-one-video-out evaluation**
   Removes the optimism of selecting and reporting on the same recordings.
4. **Fold the completed results into the manuscript**
   The IEEE draft is written with the tables generated from result files.
5. **Submit, and extend the corpus beyond 10 recordings**
   The overfitting we measured is consistent with a corpus this size.

DEMONSTRATION

What you are about to see, running live

Nothing in this demonstration is pre-rendered. The checkpoint is loaded and inference is run in front of you.

1

Load the trained checkpoint and verify the evaluation data
QT0 runs first: the rebuilt trajectory is checked against Module 4's saved windows before anything is predicted.

2

Predict a held-out moment at t+1, t+2 and t+3
DDIM sampling, 20 steps, against the naive and constant-velocity baselines on the same window.

3

Render the prediction as actual video frames
The predicted motion of 512 points is carried onto all 58,000 Gaussians and rasterised into an image.

4

Compare side by side with the real frame
Real · reconstruction ceiling · our prediction · naive baseline, with magnified motion arrows and an error map.

5

Forecast past the end of the recording
Twelve frames beyond the last recorded image — the only output that shows prediction rather than describing it.

Making the prediction visible

A prediction moves 512 points. A picture needs all 58,000. Bridging that gap is what turns a metric into something a surgeon could look at.

From points to pixels

Predicted motion of the 512 tracked points is carried onto every Gaussian by distance-weighted interpolation, so a tracked point receives its own motion exactly.

The renderer, re-verified

Module 2's renderer was re-implemented here, including degree-3 spherical harmonics. We checked it against the original rather than assuming they matched.

Six panels per moment

Real frame, reconstruction ceiling, prediction, naive baseline, motion overlay and an amplified error map — with image metrics over the full frame and over tissue alone.

Predicted motion was invisible

At 1–2 pixels, predicted and real frames look identical. Motion is drawn as arrows coloured by agreement with truth, magnified automatically — and the factor is printed on every panel, never applied to any reported number.

Nothing on screen was a prediction

Every predicted frame used to be paired with a frame that already existed. The forecast now runs 12 frames past the end of the recording: the real-video panel stops and says so, while the predicted panel keeps going.

30.04
reconstruction ceiling, dB

26.79
our prediction, dB

26.74
naive baseline, dB

The model knows which way tissue moves — not how far

Chamfer Distance alone cannot separate "learned nothing" from "learned the direction but the wrong distance". Measuring them separately can.

Agreement between predicted and true motion direction

t+1	t+2	t+3	t+5	t+10
+0.57	+0.40	+0.26	−0.01	−0.49

Why this matters

A model that learned nothing would read 0.00 everywhere.
+0.57 with naive-level error can only mean the magnitude is wrong — a far more tractable problem.

What follows from it

- Predict the correction to constant velocity (Module 5)
- Calibrate the magnitude at inference (Module 6)

Both implemented; evaluation is the next run.

The model knows when it should not be trusted

r = 0.69
spread vs actual error

5
futures sampled per input

5×
uncertainty growth over 10 frames

When the five sampled futures disagree, the prediction really is worse.

Uncertainty rises smoothly with how far ahead we predict — 0.00011 at one frame, 0.00055 at ten.

Why this is the result we are proudest of

A system that is confidently wrong is unusable in theatre.
A system that signals "do not trust me here" is usable even when imperfect.

No method in our comparison provides this.

Stated carefully

This is correlation between predicted spread and realised error. Some of it may reflect that busy moments are both harder and more variable — we say so in the paper rather than claiming more than we measured.

A two-line formula beats our model

Method	k = 1	k = 2	k = 3
Nothing moves (naive)	0.0093	0.0165	0.0222
Constant velocity	**0.0055**	**0.0138**	**0.0187**
Ours — diffusion	**0.0093**	**0.0165**	**0.0225**

Chamfer Distance, lower is better · full held-out set, 10 videos

Read this honestly

Our model sits exactly on the naive baseline at every horizon.

Constant velocity beats both — by 41% at one frame ahead.

It loses on all 10 videos.

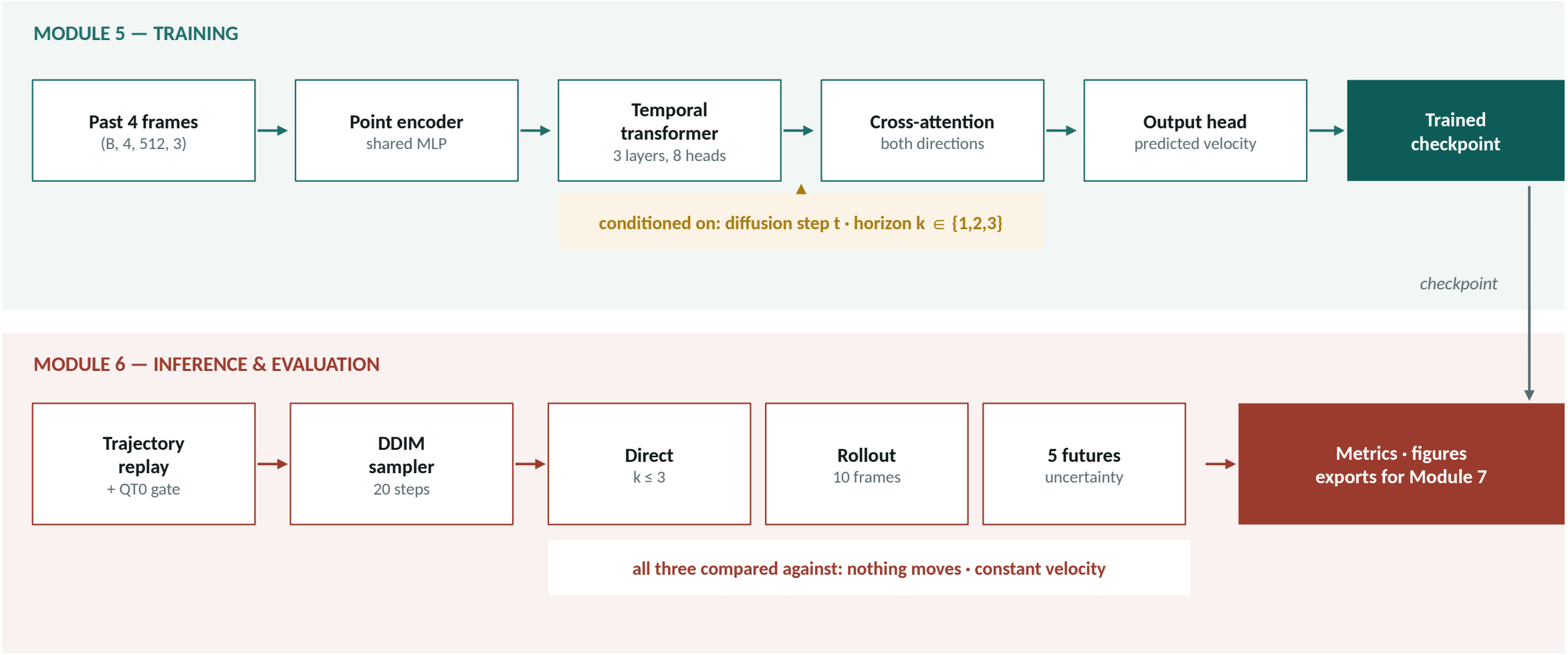
Rollout stays stable (growth exponent 0.61) and two explanations were tested and eliminated:

- Error does not grow with distance from training frames (ρ = +0.02, p = 0.61)
- Dropped preprocessing frames do not explain it (333 origins checked, 0 drops found)

Reference slides

Everything beyond this point is held for questions: the architecture, each module in detail, the problems we hit and fixed, the training curve and the literature table.

Architecture — Modules 5 and 6



Module 5 — learning how tissue moves

A conditional diffusion model that learns the deformation of 512 tracked tissue points, trained entirely without clinical labels.

Input
- Last 4 frames of 512 tracked points
- How far ahead to predict (1, 2 or 3)
- A patch of random noise to shape

Model
- 16.3 M parameters
- Temporal transformer per point
- Two-way cross-attention
- Denoises over 20 steps

Output
- Movement of each point
- Sample repeatedly → several plausible futures

Why diffusion and not direct regression
Soft tissue under contact is genuinely uncertain. A single estimate hides that. Sampling the model repeatedly gives a spread of outcomes — which Module 6 shows is a usable confidence signal.

Why the horizon is an input
The same 4 frames appear in training with targets 1, 2 and 3 frames ahead — three different right answers. Without being told which, the task is unlearnable no matter the capacity.

Built from 7,371 windows across 10 reconstructed videos — 5,910 train, 1,461 validation — with no human annotation at any stage.

Module 5 — what went wrong, and what we did

The task had no single right answer One context appeared with targets 1, 2 and 3 frames ahead.	→	**Condition on the horizon** Horizon embedded and injected into both branches of the network.
The signal was smaller than the noise Tissue moves 0.004 units; diffusion adds noise at scale 1.	→	**Rescale the target to unit variance** Inverted before every reported metric, so results stay in physical units.
Raw trajectories had almost no momentum Correlation between consecutive motions was 0.077 — nearly noise.	→	**Adaptive temporal smoothing** Per-video Savitzky–Golay window; correlation rose to 0.89.
Training made the model worse after epoch 60 Held-out error rose while training loss kept falling.	→	**Shorter schedule, stronger regularisation** 500 → 200 epochs, weight decay 0.01 → 0.05, best checkpoint kept.
A model that gave up scored the same as doing nothing Predicting zero displacement equalled the naive baseline.	→	**Predict the correction to constant velocity** Now giving up falls back to an estimator 41% better than naive.

Module 6 — using the model, and measuring it honestly

Module 6 trains nothing. It loads the trained model, asks it three different questions, and grades every answer against estimators that need no learning at all.

Ask once

Where will tissue be 1, 2 or 3 frames from now?

Most reliable mode.

Ask repeatedly

Feed each answer back in to reach 10 frames ahead.

Error compounds — measured.

Ask five times

Five possible futures from one input. How much they disagree is the confidence estimate.

Graded against two simple rules

- "Nothing moves" — the naive baseline
- "Everything continues straight" — constant velocity

If a 16 M-parameter model cannot beat these, we say so.

Six quality tests

QT0 verifies the evaluation data is the data the model trained on, before anything is predicted. QT1–QT5 cover accuracy, rollout, diversity, smoothness and cross-scene generalisation.

Module 6 is not there to make the model look good. It is there to establish, with evidence, exactly where the model stands.

Module 6 — what went wrong, and what we did

Evaluation could have measured the wrong data Rollout needs continuous trajectories; Module 4 stores fixed windows.	→	**Rebuild, then verify before predicting** QT0 checks the rebuild against Module 4 output and runs as a gate.
The metric was penalising the model unfairly Chamfer Distance is minimised by the average, not by one sample.	→	**Average several sampled trajectories** Removes a ~40% penalty the model was being charged for free.
Diagnostics were measuring one easy video Taking the first N windows drew almost entirely from one recording.	→	**Evenly spread the evaluation set** Every video contributes; earlier scores had been 4× too optimistic.
Our constant-acceleration baseline was wrong A backward difference gives the velocity at t − ½, not at t.	→	**Derive the velocity at t properly** $v = (3 \cdot x(t) - 4 \cdot x(t-1) + x(t-2)) / 2$, verified exact on quadratic motion.
What we called a Kalman filter was not one It was an alpha–beta filter with fixed gains and no covariance.	→	**Implement a real two-state Kalman filter** Full 2×2 covariance; recovers a known velocity to 2.6e−06.

Training converged cleanly — and then overfitted

16.3 M
parameters

500
epochs completed

0.012
training loss, from 0.36

~60
best epoch

What went right

Smooth convergence, no instability, no divergence.
Multi-session training resumes exactly where it stopped,
surviving Colab disconnection.

What went wrong

Held-out error bottomed out near epoch 60 and rose after
~260 while training loss kept falling — overfitting,
measured directly rather than assumed.

Best held-out Chamfer Distance 0.01568 versus naive baseline 0.01557

The model converged to predicting almost no motion — which is exactly what the naive baseline does. Understanding why is the rest of this presentation.

Four adjacent fields. None occupies the cell we need

Line of work	Representative	Predicts future?	Output space	Uncertainty
Surgical 4D reconstruction	EndoNeRF, LerPlane, EndoGaussian, Deform3DGS	No — retrospective	3D geometry	No
Surgical video anticipation	TPG-VAE, SurgVista	Yes	2D pixels only	No
Point cloud forecasting	PointRNN, MoNet, PCPNet	Yes	3D, rigid outdoor LiDAR	No
Medical point diffusion	Xie et al. 2025 (liver)	No — present state	3D points	No
This work	**—**	**Yes**	**3D tissue geometry**	**Calibrated**

The contribution is the combination in the last row, not any single column: future 3D tissue geometry, from monocular video, with no annotation, and a calibrated confidence estimate.

WHERE WE ARE

FYP-1 built the reconstruction. FYP-2 adds the prediction

M1	M2	M3	M4	M5	M6
Frames + masks	4D Gaussian reconstruction	Trajectory extraction	Self-supervised windows	Diffusion predictor	Inference + evaluation
FYP-1 complete	FYP-1 complete	FYP-1 complete	FYP-1 complete	FYP-2 focus	FYP-2 focus

30.10 dB
reconstruction PSNR

0.904
reconstruction SSIM

57,943
Gaussians tracked

7,371
windows built (train + val)

The laparoscope shows the present, never the next moment

A surgeon operates through one camera. It gives a flat image with no depth, no 3D structure, and no indication of how tissue will respond to the instrument about to touch it.

Reconstruction research has solved the first problem well. It has not addressed the second at all.

Every published surgical 4D reconstruction system is retrospective: it models deformation that has already been recorded.

What exists today
Reconstruct the scene in 3D, for frames already captured

What is missing
Anticipate where tissue will be a moment from now, with an honest indication of confidence

0	1–2 px	0	10
depth sensors available	tissue motion per frame	deformation labels in clinical video	videos reconstructed

Where Modules 5 and 6 stand

Delivered

- A complete prediction pipeline, no clinical labels
- A six-test evaluation framework with an integrity gate
- Calibrated uncertainty (r = 0.69)
- Comparative study implemented (11 methods)
- Conference paper drafted (IEEE format)

Not yet achieved

- Beating the naive baseline on accuracy

We localised the cause to displacement magnitude, not direction, and implemented two fixes that follow directly from that measurement. Evaluating them is the immediate next step.

An honest negative result with an identified cause is worth more than an unexplained positive one.